



Mixture of Noise for Pre-Trained Model-Based Class-Incremental Learning

Hanna Hamrell - CV Reading Group @ RPL
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Mixture of Noise for Pre-Trained Model-Based Class-Incremental Learning

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Abstract

Class Incremental Learning (CIL) aims to continuously learn new categories while retaining the knowledge of old ones. Pre-trained models (PTMs) show promising capabilities in CIL. However, existing approaches that apply lightweight fine-tuning to backbones still induce parameter drift, thereby compromising the generalization

Jiang, K., Shi, Z., Zhang, D., Zhang, H., Li, X. (2025). **Mixture of Noise for Pre-Trained Model-Based Class-Incremental Learning**. In: *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. San Diego, CA. <https://openreview.net/forum?id=wI6oHXeTR8>



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- Why well made use of noise might help with catastrophic forgetting in class-incremental learning (CIL) for pre-trained models (PTMs)



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- Experiments with π -noise in visual transformers evaluated on standard benchmarking datasets
- Well placed π -noise created a new SOTA method for analytic CIL



The challenge with continual learning for pre-trained models

The accuracy of PTMs degrade for additional tasks due to the parameter drift, which eventually causes catastrophic forgetting.

Not all noise is harmful

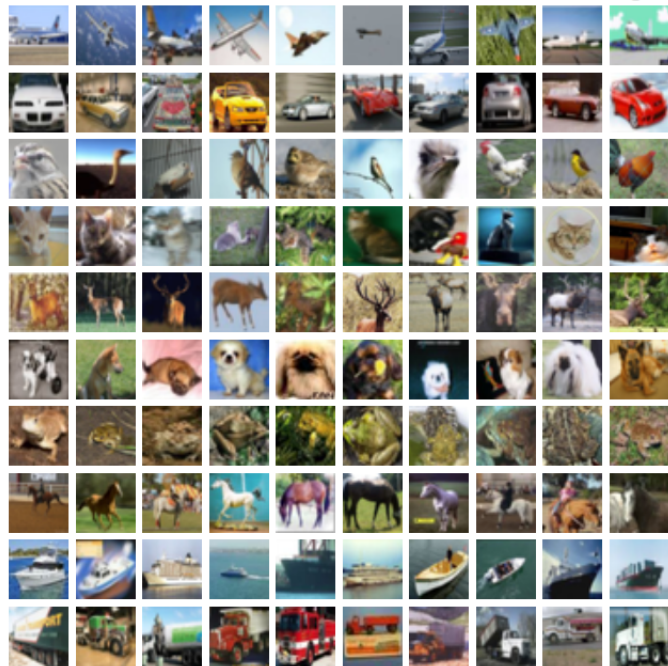
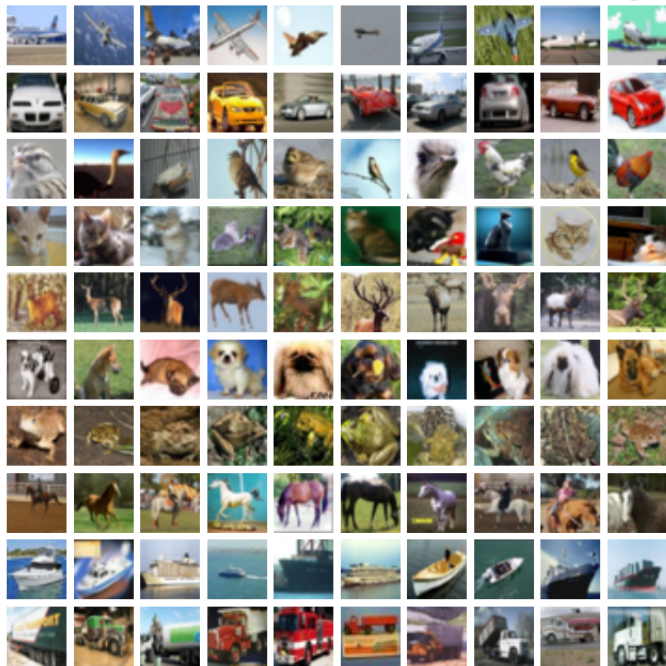


Image sources: <https://cave.cs.toronto.edu/kriz/cifar.html>

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Not all noise is harmful



Xuelong Li, "Positive-incentive Noise," *IEEE Transactions on Neural Networks and Learning Systems*, DOI: 10.1109/TNNLS.2022.3224577.

1

Positive-incentive Noise

Xuelong Li, *Fellow, IEEE*

Abstract—Noise is conventionally viewed as a severe problem in diverse fields, e.g., engineering, learning systems. However, this paper aims to investigate whether the conventional proposition always holds. It begins with the definition of task entropy, which extends from the information entropy and measures the complexity of the task. After introducing the task entropy, the noise can be classified into two kinds, Positive-incentive noise (P_i -noise or π -noise) and pure noise, according to whether the noise can reduce the complexity of the task. Interestingly, as shown theoretically and empirically, even the simple random noise can be the π -noise that simplifies the task. π -noise offers new explanations for some models and provides a new principle for some fields, such as multi-task learning, adversarial training, etc. Moreover, it reminds us to rethink the investigation of noises.

acoustics tasks. In most scenes, the car noises should be regarded as useless information or disturbance due to the unsatisfied reception when collecting data. However, if the acoustics task is relevant to time, the car noise may offer extra information about time and *enhance* performance. Generally speaking, the intensities of car noises in the morning rush hour and midnight are clearly different, which can provide coarse information about time. Another inspiring instance is the *gum example*. For a clean wall, either gum or nail is a kind of noise for the wall. It means that the gum stuck on the wall and embedded nail are both unexpected. However, a piece of gum may help to remove the embedded nail with the help of

Image sources: X. Li, "Positive-Incentive Noise," in *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 6, pp. 8708-8714, June 2024, doi: 10.1109/TNNLS.2022.3224577.



Learning beneficial noise is possible

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IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE

Variational Positive-incentive Noise: How Noise Benefits Models

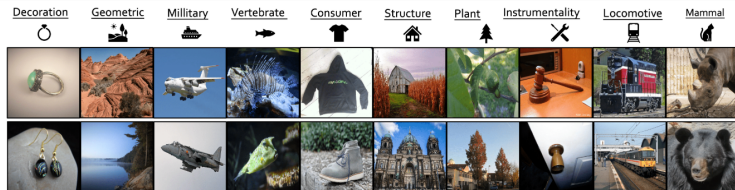
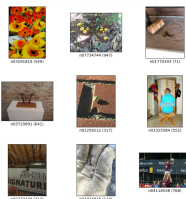
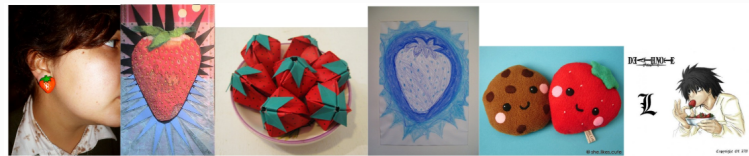
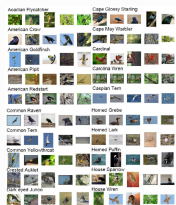
Hongyuan Zhang, Sida Huang, Yubin Guo, and Xuelong Li*, *Fellow, IEEE*

Abstract—A large number of works aim to alleviate the impact of noise due to an underlying conventional assumption of the negative role of noise. However, some existing works show that the assumption does not always hold. In this paper, we investigate how to benefit the classical models by random noise under the framework of Positive-incentive Noise (*Pi-Noise*) [1]. Since the ideal objective of *Pi-Noise* is intractable, we propose to optimize its variational bound instead, namely variational *Pi-Noise* (*VPN*). With the variational inference, a *VPN* generator implemented by neural networks is designed for enhancing base models and simplifying the inference of base models, without changing the architecture of base models. Benefiting from the independent design of base models and *VPN* generators, the *VPN* generator can work with most existing models. From the extensive experiments on different base models (including linear models, ResNet, ViT, *etc.*) it is shown that the proposed *VPN* generator can improve the base models. It is appealing that the trained *VPN* generator prefers to blur the irrelevant ingredients in complicated images, which meets our expectations.

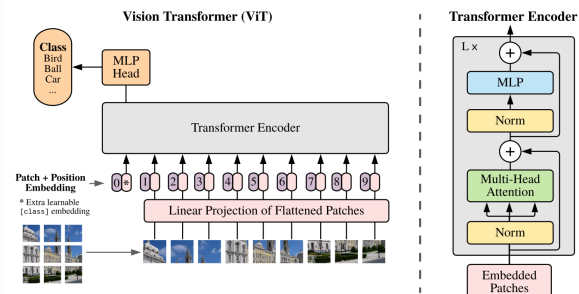
Index Terms—Positive-incentive Noise, Beneficial Noise, Variational Inference.

Image sources: H. Zhang, S. Huang, Y. Guo and X. Li, "Variational Positive-Incentive Noise: How Noise Benefits Models," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 47, no. 9, pp. 8313-8320, Sept. 2025, doi: 10.1109/TPAMI.2025.3575295.

Scenario



List of image sources to be found on final slide.



A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," arXiv preprint arXiv:2010.11929, 2021. [Online]. Available: <https://arxiv.org/abs/2010.11929>



Proposed solution and main contributions

- Frozen PTM as base combined with an analytical classifier for CIL.



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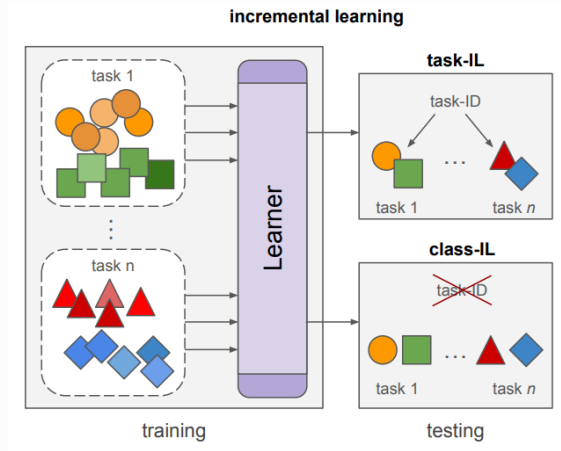
- Frozen PTM as base combined with an analytical classifier for CIL.
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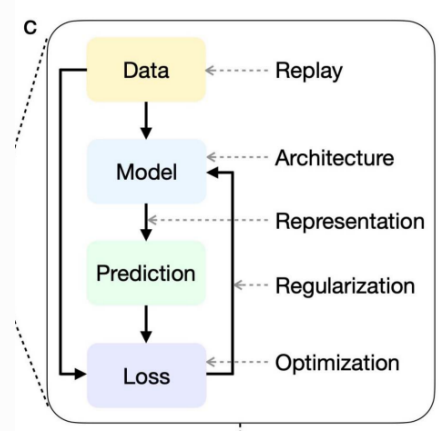
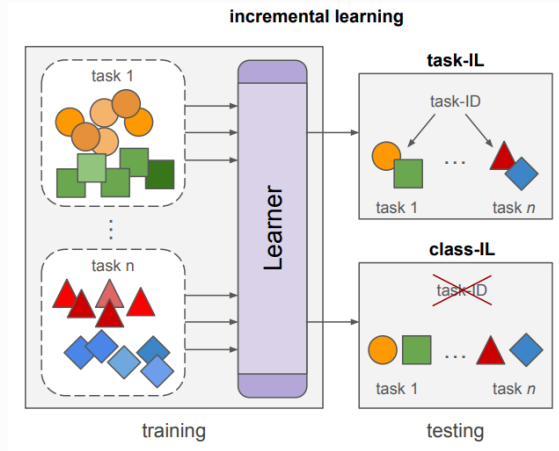
- Frozen PTM as base combined with an analytical classifier for CIL.
- To learn π -noise for each new task, and add it to intermediate model features, to suppress non-essential features.
- To adaptively learn how to mix the per-task noises at inference.

Class Incremental Learning



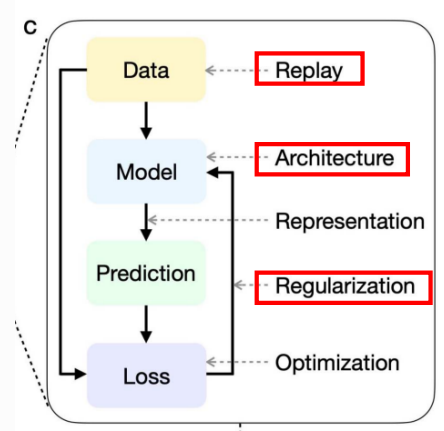
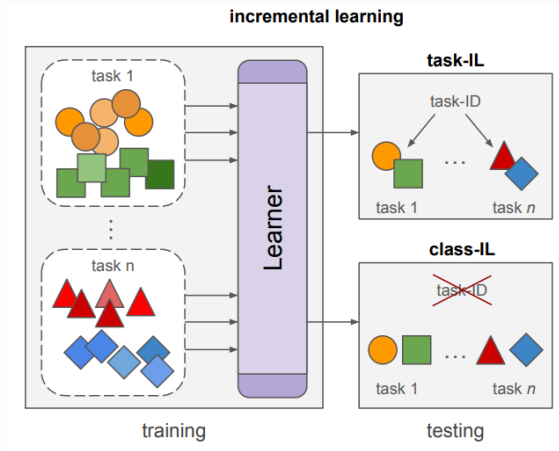
Left: image modified from arxiv.org/pdf/2010.15277 Right: image from [ieeexplore.ieee.org \(arnumber=10444954\)](https://ieeexplore.ieee.org/arnumber=10444954)

Class Incremental Learning



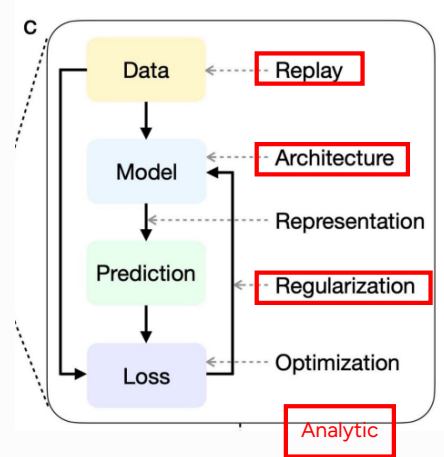
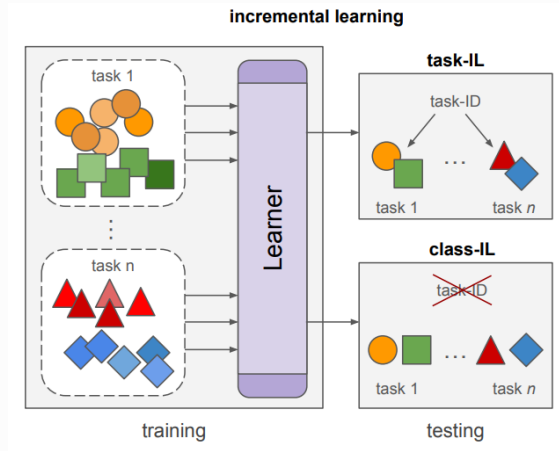
Left: image modified from arxiv.org/pdf/2010.15277 Right: image from [ieeexplore.ieee.org \(arnumber=10444954\)](https://ieeexplore.ieee.org/arnumber=10444954)

Class Incremental Learning



Left: image modified from arxiv.org/pdf/2010.15277 Right: image from [ieeexplore.ieee.org \(arnumber=10444954\)](https://ieeexplore.ieee.org/arnumber=10444954)

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Pre-trained model (PTM) based CIL

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- When we take a frozen or near-frozen PTM and limit adaptation to small things, e.g. prompts, extra branches, feature statistics etc.
- MIN keeps the backbone PTM frozen, but adds beneficial noise to intermediate features.



Analytic learning

Iterative solving of classifier weights through linear regression over frozen backbone features. Let $D = \{X, Y\}$ be the dataset with inputs X and labels Y . Let W be classifier weights and R a fixed-sized correlation matrix.



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Can be rewritten on iterative form as

$$W_t = [W_{t-1} - R_t \mathcal{F}(X_t) W_{t-1}, R_t \mathcal{F}(X_t)^\top Y_T].$$



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Let t be a task, H be task entropy, and ϵ be noise. π -noise is the noise that satisfies the condition

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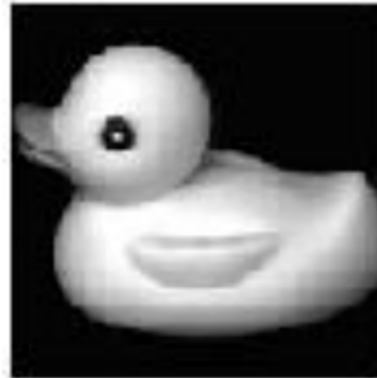
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Examples of π -noise

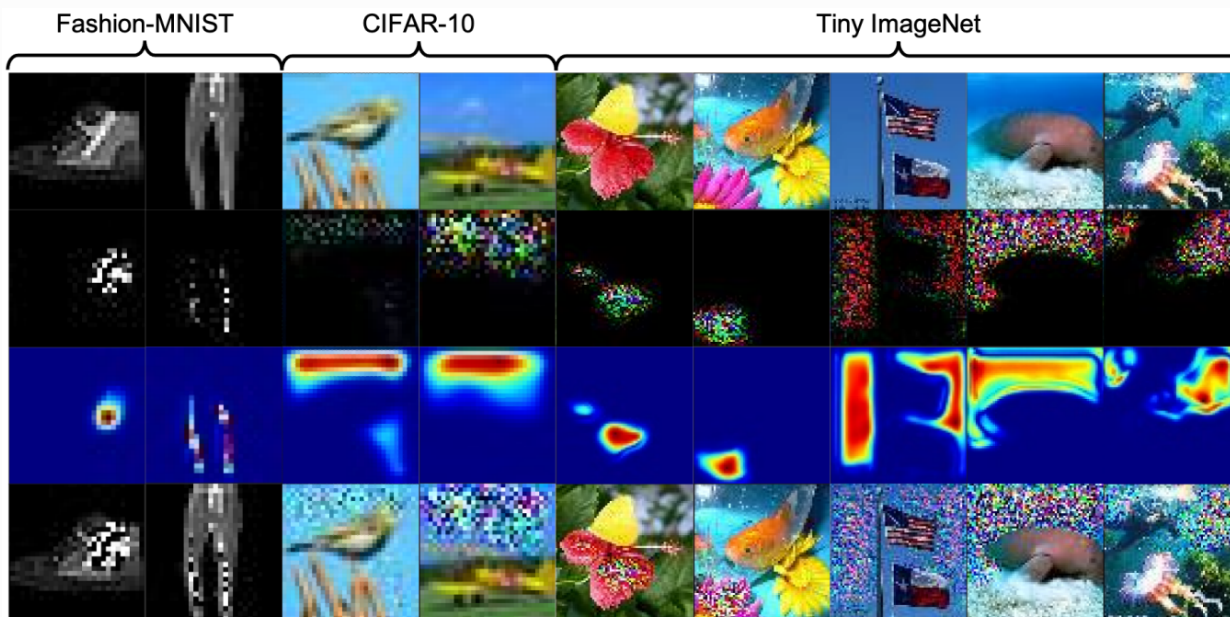
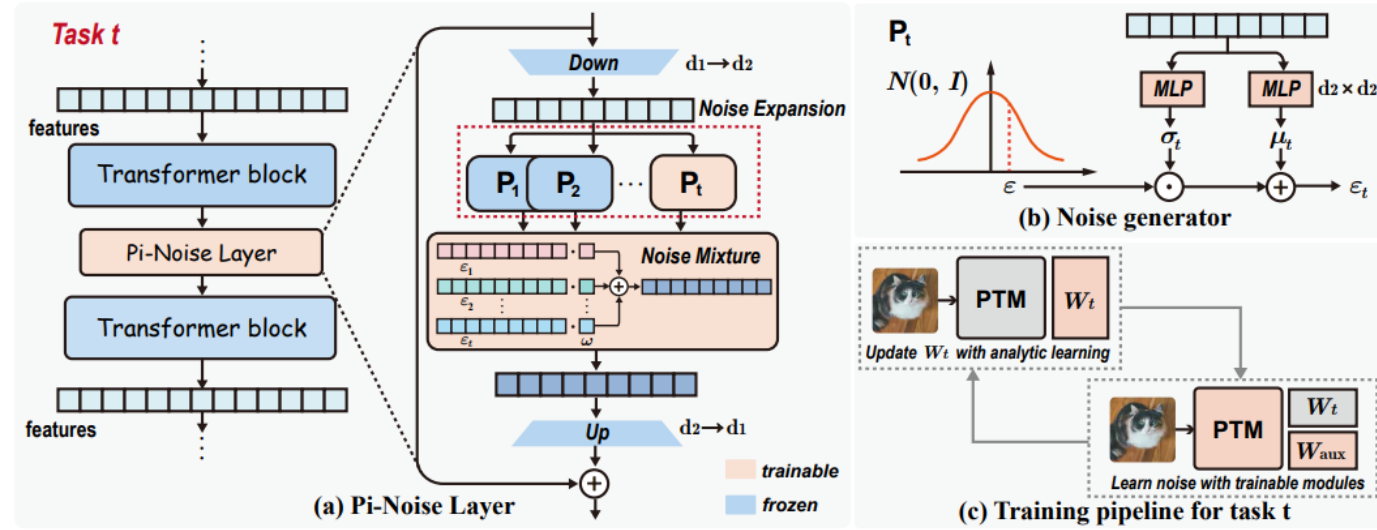
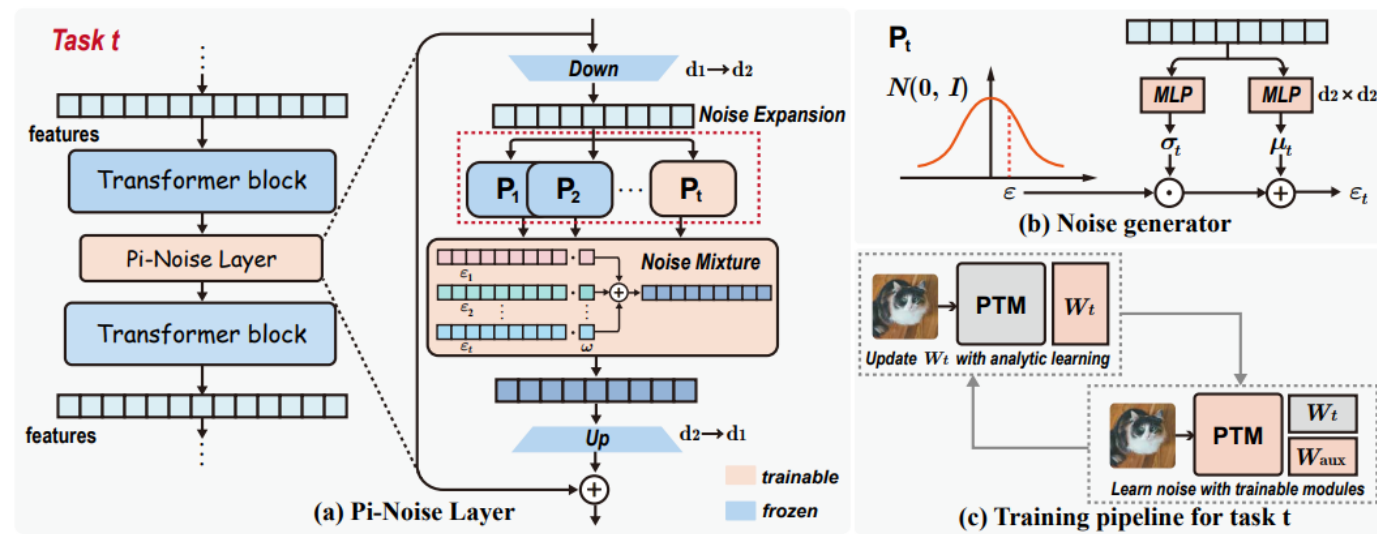


Image sources: H. Zhang, S. Huang, Y. Guo and X. Li, "Variational Positive-Incentive Noise: How Noise Benefits Models," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 47, no. 9, pp. 8313-8320, Sept. 2025, doi: 10.1109/TPAMI.2025.3575295.

Proposed approach



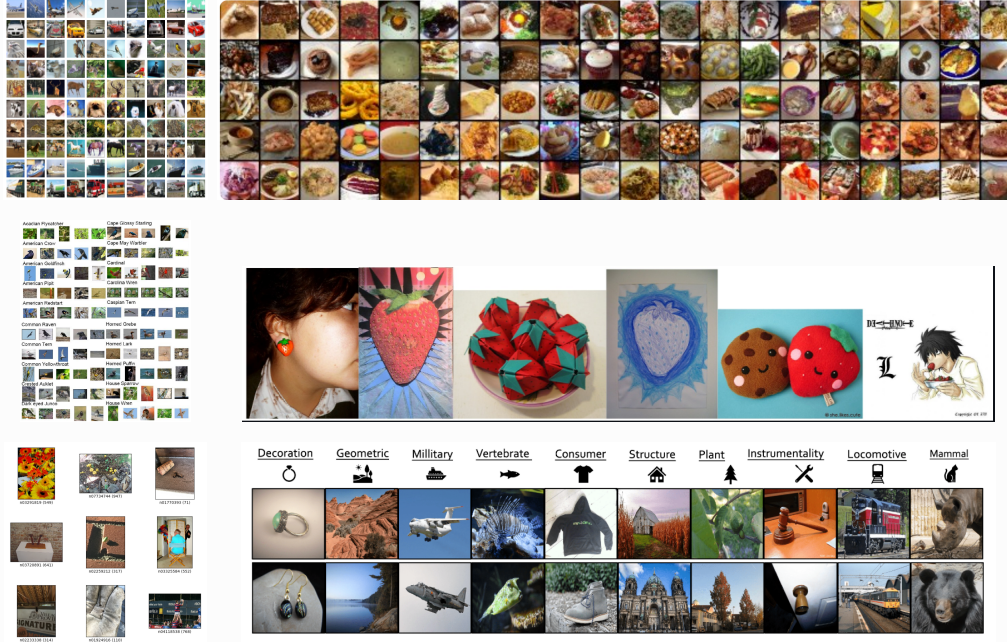
Proposed approach



$$\mathcal{L}_{cls} = l_{CE}(z_L W_{aux}, y - z_L W_t)$$

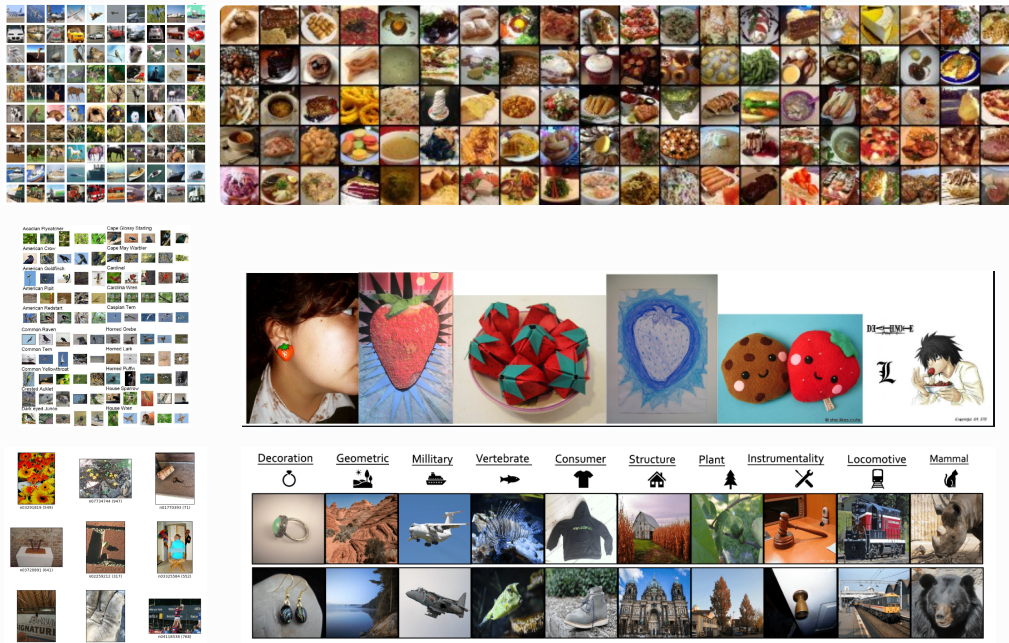
(1)

Experimental setup



List of image sources to be found on final slide.

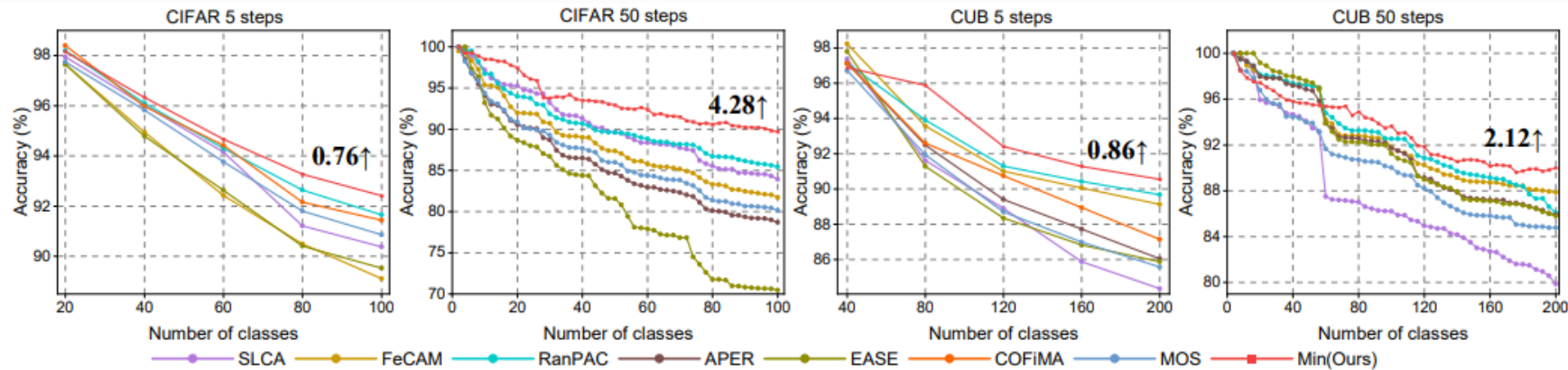
Experimental setup



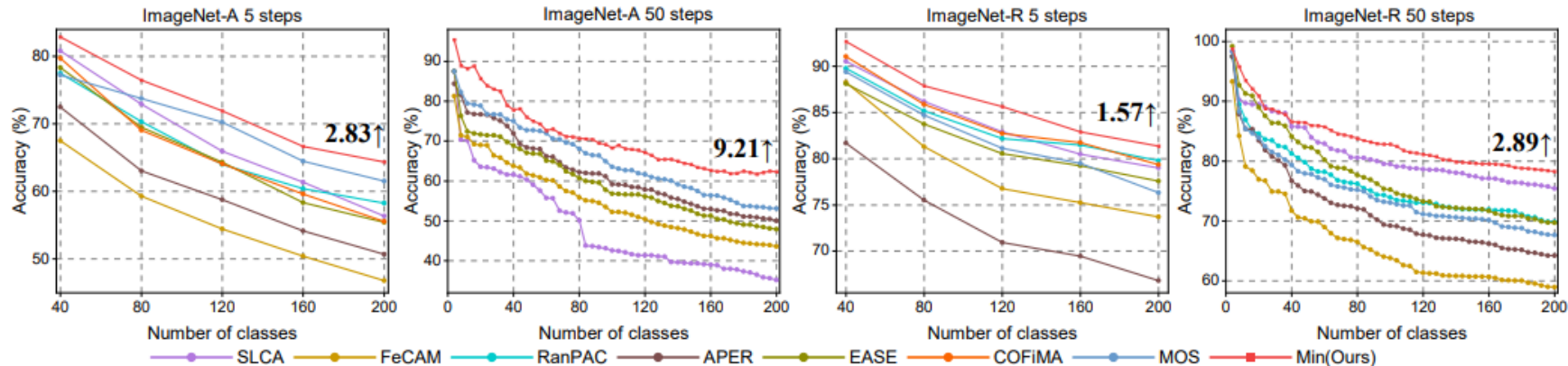
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Split each dataset into $\mathcal{T} = 5, 10, 20$ or 50 tasks.

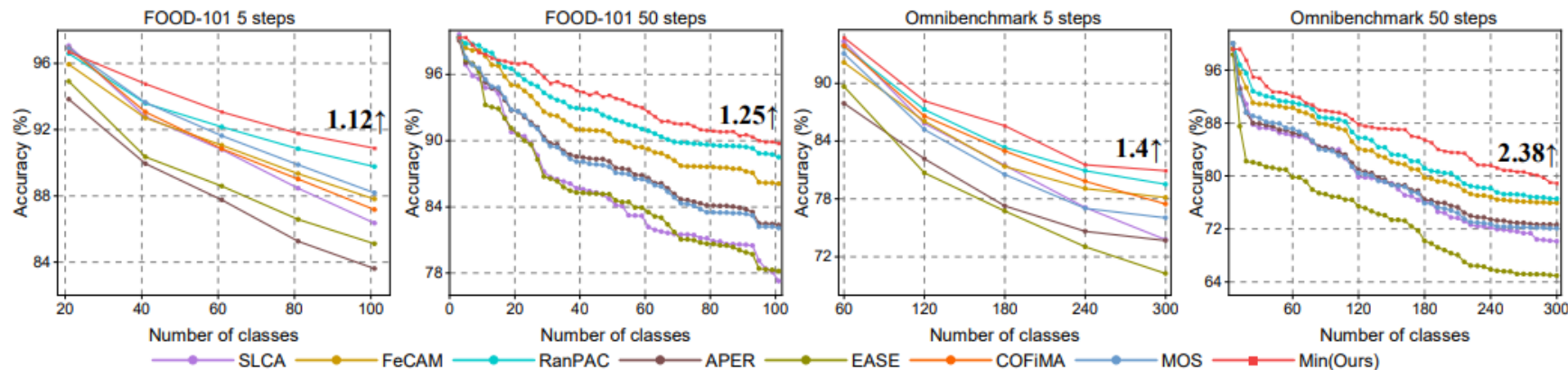
Comparison with SOTA on CIFAR-100 and CUB200



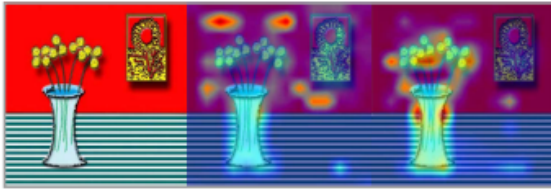
Comparison with SOTA on ImageNet-A and ImageNet-R



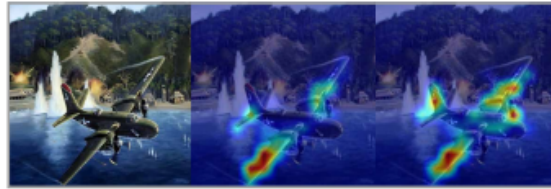
Comparison with SOTA on FOOD-101 and Omnibenchmark



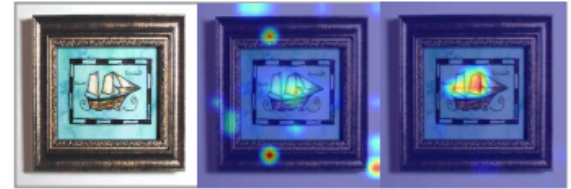
Baseline vs MIN Grad-CAM visualization



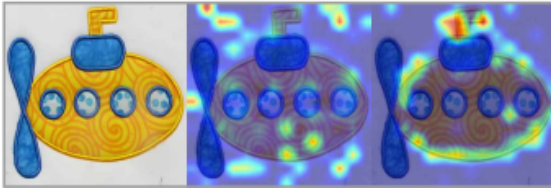
vase



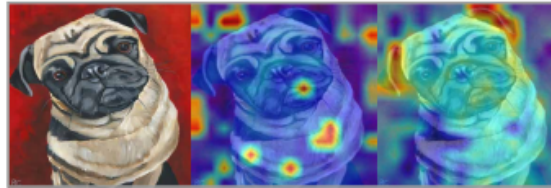
warplane



schooner



submarine



pug



bald eagle

Conclusions

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- The addition of the π -noise makes the classifier rest on genuinely discriminative decision boundaries, making the job of the analytic classifier easier, thus mitigating forgetting.
- The MIN method proves effective in the evaluated CIL scenarios using a ViT backbone.





Additional image references

From left to right, from top to bottom:

<https://cave.cs.toronto.edu/kriz/cifar.html>

https://data.vision.ee.ethz.ch/cvl/datasets_extra/food-101/

<https://www.semanticscholar.org/paper/The-Caltech-UCSD-Birds-200-2011-Dataset-Wah-Branson/c069629a51f6c1c301eb20ed77bc6b586c24ce32/figure/0>

<https://github.com/hendrycks/imagenet-r>

[https://www.tensorflow.org/datasets/catalog/imagenet_α](https://www.tensorflow.org/datasets/catalog/imagenet_alpha)

<https://zhangyuanhan-ai.github.io/OmniBenchmark/>